11. (a) $^{226}_{88}\text{Ra} \rightarrow ^{222}_{86}\text{Rn} + ^{4}_{2}\text{He}$ / He / α 2A 2

(b) $\Delta m = 226.0254 - (222.0176 + 4.0026) = 0.0052$ u 1M

Energy released = (0.0052)(931) = 4.84 (MeV) 1A 2

(c) Number of radium atoms in the source

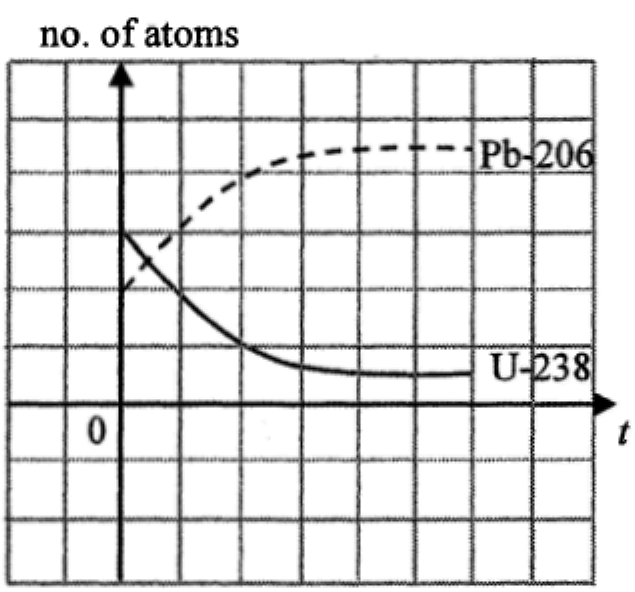
$$N = N_A\left(\frac{1}{226}\right) \times (5 \times 10^{-6}) = (6.02 \times 10^{23})\frac{1}{226} \times (5 \times 10^{-6}) = 1.33 \times 10^{16}$$ 1A

Activity $A = \dfrac{\ln 2}{t_{1/2}} \cdot N$ 1M

$$= \frac{\ln 2}{1600 \times 365 \times 24 \times 3600} \cdot 1.33 \times 10^{16}$$

$= 1.83 \times 10^5$ (disintegrations per second, Bq) 1A 3

10. (a) Mass deficit $= (2.014102 + 3.016049)\text{ u} - (4.002602 + 1.008665)\text{ u}$ $= 0.018884\text{ u}$	1M		
Energy released $= 0.018884 \times 931$ MeV $= 17.58$ (MeV)	1A		
<u>Or</u> Energy released $= 0.018884 \times 1.661 \times 10^{-27} \times c^2$ $= 2.823 \times 10^{-12}$ J or 17.64 MeV	1A		
		2	
(b) (i) To overcome the (electrostatic) repulsion between the two (positive) nuclei and	1A		
becomes electrical potential energy (of the two nuclei).	1A		
		2	
(ii) High temperature enables them to have sufficient kinetic energy (to overcome electrical repulsion between the nuclei).	1A		
		1	
(iii) Kinetic energy becomes electrical potential energy $E_p = 2 \times \frac{1}{2} m\,(c_{rms})^2 = 2 \times \frac{3RT}{2N_A}$	1M		Accept without the factor '×2'
0.4 MeV $= 2 \times (\frac{3 \times 8.31 \times T}{2 \times 6.02 \times 10^{23}})$			0.4 MeV $= 6.4 \times 10^{-14}$ J
$T = 1.545 \times 10^9$ K i.e. order of magnitude 10^9 (K)	1A		
Alternatively: $E_p = \frac{1}{4\pi\varepsilon_0} \cdot \frac{e^2}{10^{-15}} = 2 \times \frac{3RT}{2N_A}$	1M		
$T = 5.56 \times 10^9$ K i.e. order of magnitude 10^9 (K)	1A		
		2	

9. (a) $(4)n_\alpha = 238-206 \Rightarrow n_\alpha = 8$	1A	
$(2)n_\alpha + (-1)n_\beta = 92-82 \Rightarrow n_\beta = 6$	1A	
	2	
(b) (i) $N = N_0\left(\frac{1}{2}\right)^{t/T_{1/2}}$	1M	
$\frac{3}{5} = \left(\frac{1}{2}\right)^{t/4.5\times10^9\ \mathrm{yr}}$		
Or $N = N_0\ e^{-\lambda t}$ and $\lambda = \frac{\ln 2}{T_{1/2}}$	1M	
$\therefore t = 3.316\times10^9$ years $\approx 3.3\times10^9$ years	1A	
	2	
(ii) Answer in (i) is an underestimate. The original number of U-238 atoms should be greater.	1A	Accept 'more U decayed'.
$\therefore$ the ratio $\frac{\text{present number of U}-238\text{ atoms}}{\text{original number of U}-238\text{ atoms}} = \frac{N_t}{N_0}$ is in fact smaller (than $\frac{3}{5}$), thus longer time should have been elapsed.	1A	
	2	
(iii) 	2A	
	2	

10. (a)	${}^{210}_{84}\text{Po} \rightarrow {}^{206}_{82}\text{Pb} + {}^{4}_{2}\text{He}$	2A	
			2
(b)	The α-particles ionize the air particles. The ions neutralize the charges on the dust/photo or film surface.	1A 1A	
			2
(c)	This is because α-particles have a range of only a few centimeters in air.	1A	
			1
(d)	Activity after 1 year = $\left(\frac{1}{2}\right)^{\frac{365}{138}}$	1M	
	= 0.160 unit	1A	
	Alternative method: $A = A_0 e^{-\frac{ln2}{t_{1/2}}t}$ $= 1 \times e^{-\frac{ln2}{138}(365)}$ $= 0.160$ unit	 1M 1A	
			2

Solution	Marks	Remarks
10. (a) (i) $x = 3$	1A	
	1	
(ii) More neutrons are produced in each fission for triggering further fissions, i.e. $x > 1$.	1A	
	1	
(b) (i) $m = m_0 e^{-kt}$ $k = \frac{\ln 2}{t_{1/2}} (= 9.846\times10^{-10}\ \text{yr}^{-1})$ $0.06 = m_0\, e^{-\ln 2 \times \left[\frac{2\times10^9}{7.04\times10^8}\right]}$	1M	OR $0.06 = m_0\ (\frac{1}{2})^{2\times10^9 / 7.04\times10^8}$
$m_0 = 0.429882832\ (\text{kg}) \approx 0.430\ (\text{kg})$	1A	
	2	
(ii) $\frac{0.430}{13.556 + 0.430} = 0.03073691 \approx 3.1\,\% > 3\%$ Thus natural nuclear fission was possible.	1M/1A	
	1	
(c) Underground water might run dry.	1A	
OR Energy released by fission drys up the underground water.	1A	
Therefore, fission might stop without slow neutrons.	1A	
	2	

10. (a) proton / ^1_1H / p / hydrogen nucleus	1A	Accept: proton / ^1_1H / p / ^1_1p / H / hydrogen nucleus / hydrogen / hydrogen atom NOT accept: H_2
	1	
(b) Change in mass $\Delta m = (16.9947 + 1.0073) - (13.9993 + 4.0015)$ $= 0.0012$ u	1M	1M for Δm
Energy $= 0.0012 \times 931$ $= 1.1172$ (MeV) ≈ 1.12 (MeV) ~~(Or = $0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$~~ ~~$= 1.79388 \times 10^{-13}$ J $\approx 1.79 \times 10^{-19}$ MJ)~~	1A	Accept 1.10 to 1.12 (MeV) Answer should be in MeV, not J. MeV could be omitted [Energy $= 0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ $= 1.79388 \times 10^{-13}$ J $\approx 1.79 \times 10^{-19}$ MJ ≈ 1.12 (MeV)] ~~Accept $(1.79$ to $1.80) \times 10^{-19}$ MJ~~
	2	
(c) By the conservation of momentum, as before the reaction occurs the α particle has momentum, the total momentum of the products (= momentum of the α particle) must also be non-zero,	1A	1st 1A should mention the product(s) has/have momentum
i.e. the total KE of the products must greater than 0, thus the α particle should have a larger KE.	1A	2nd 1A KE/speed of the products must greater than zero Accept: kinetic energy shared by other particles (NOT accept :energy shared.) NOT accept: against repulsion of nucleus / inelastic collisions / KE loss / consider binding energy [The kinetic energy of the α particle = energy calculated in (b) + total KE of the products]
	2	

10.	(a)	Decrease in mass in the reaction (for two ${}^{2}_{1}\text{H}$) $\Delta m = (2 \times 2.014102) - (3.016029 + 1.008665)$ $= 4.028204 - 4.024694 = 0.003510\ \text{u}$ Maximum energy released by 1 mole of hydrogen nuclides $= \left(\dfrac{6.02 \times 10^{23}}{6420 \times 2}\right) \times (0.003510 \times 931\ \text{MeV})$ $= 1.532104 \times 10^{20}\ (\text{MeV}) \approx 1.53 \times 10^{20}\ (\text{MeV})$	1M 1M 1A 3	
	(b)	${}^{2}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow \underline{{}^{3}_{1}\text{H} + {}^{1}_{1}\text{H}}$ (or $\underline{{}^{1}_{1}\text{p}}$)	1A 1	
	(c)	Any TWO: - Unlike fission, the fuel (hydrogen) for fusion is abundant and widely available. - Unlike fission, the products of fusion reactions are not radioactive. - Much larger amount of energy is produced by fusion (for equal mass of fuel).	2A 2	

9.	(a)	(i)	alpha particle / α	1A	
				1	
		(ii)	6 α-decays, 4 β-decays $4\,n_\alpha = 232 - 208$ $\quad n_\alpha = 6$ $2\,n_\alpha + (-1)\,n_\beta = 90 - 82$ $\quad n_\beta = 4$	 1A 1A	
				2	
	(b)		$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} = \left(\frac{1}{2}\right)^{\frac{10}{1.41\times10^{10}}}$ Portion $= 1 - \frac{N}{N_0} = 1 - \left(\frac{1}{2}\right)^{\frac{10}{1.41\times10^{10}}}$ $= 4.9159 \times 10^{-10} \approx 4.9 \times 10^{-10}$ (2 sig. fig.)	1M 1A	$N = N_0\,e^{-\lambda t}$ and $\lambda = \frac{\ln 2}{t_{1/2}}$ $\lambda = \frac{\ln 2}{t_{1/2}} = 4.9159 \times 10^{-11}\ \text{yr}^{-1}$ $1 - \frac{N}{N_0} = 1 - e^{-\frac{\ln 2}{t_{1/2}}t}$ $= 1 - e^{-\frac{\ln 2}{1.41\times10^{10}}(10)}$ $\approx 4.9 \times 10^{-10}$
				2	
	(c)	(i)	Aluminium body blocks alpha and/or beta particles emitted from thoriated glass. Or Short range of α particles	1A 1A	
				1	
		(ii)	Thorium-232 has an extremely long half-life. Or As part (b) suggests, only an extremely small fraction (4.9×10^{-10}) of thorium decaying in 10 years.	1A 1A	
				1	